

Summary of Krizhevsky et. al.'s 2012 paper :

ImageNet Classification with Deep Convolutional Neural Networks

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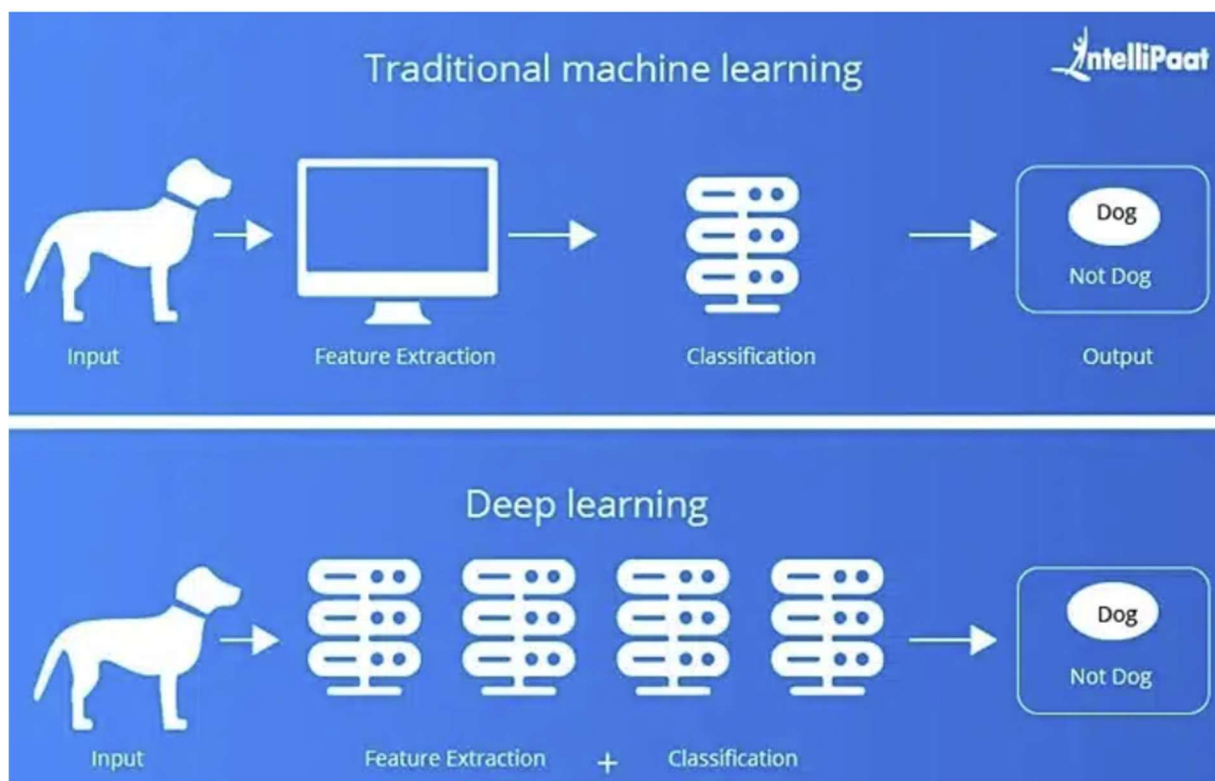


Image source: IntelliPaat

Introduction:

The Krizhevsky et. al.'s 2012 paper discusses the results of a machine learning competition held in 2012, for which the authors used a CNN that produced, for that time, surprisingly good results, the best in the contest. Their findings have since become the standard for machine image recognition.

Procedures:

Problem setting: In order to make object recognition work in real-world environments, we need large datasets. And in order to make machine learning work effectively on large datasets, this paper demonstrates that deep CNNs are the way to go.

The architecture they settled on (five convolutional layers followed by three fully-connected layers) was validated experimentally. They found that removing any convolutional layer degraded performance, suggesting that depth is essential.

The model was further optimized with the following techniques:

- ReLU (replacing the previously standard tanh activation function) made training significantly faster
- Parallel GPUs reduced training time
- Normalization improved generalization
- Overlapping pooling : pooling summarizes a local region of a feature map into a single value. Normally the pooling windows don't overlap. The authors found that using slightly larger windows that overlap between steps (the steps itself don't change) reduces overfitting and marginally improves accuracy.

In order to combat overfitting, the authors also proposed:

- Data augmentation: dynamically creating slightly modified versions of existing images during training
- Dropout: at each batch, a random set of neurons is switched off, preventing the model from becoming too dependent on any particular neuron

Results:

The CNN model proposed by the authors won the 2012 ImageNet competition with a top-5 error rate of 15.3%, compared to 26.2% for the second-best entry, which used traditional hand-crafted features rather than a CNN. A gap of nearly 11 percentage points, which is remarkable for a machine learning competition.

Conclusion:

The authors conclude that deep CNNs trained on large datasets are clearly superior to traditional approaches for image recognition. They also note that their results were likely limited by available computing power at the time. A bigger network trained on more data would probably do even better.

Personal Notes:

Being new to machine learning, I learned the concepts described in this paper as standard practice - the way things are done, and apparently have always been done. Reading this paper was a reminder that "always" means 2012 in this field! Things move fast in machine learning.

The second thing that strikes me is more philosophical. The previous approach to image recognition was based on human expertise : specialists manually describing what to look for in an image. CNN replaced that with pure mathematics: multiplications, additions, and derivatives, repeated millions of times, with no human telling the system what a cat or a dog looks like. The fact that the mathematical approach wins is counterintuitive. It suggests that in some domains, we are better off letting the math figure it out than trying to encode human knowledge explicitly.